

STAT5102

Regression in Practice

Group Project

Group 8

Topic:

Predicting the severity of accidents
based on traffic conditions

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1. Introduction

Hong Kong is a compact metropolitan, characterizing one of the most intense, densely packed environments in the world. With over 7.5 million people living on around 24% of the developed land, road and vehicle usage is highly concentrated (Intelligent Transport Systems (ITS) - Introduction, 2025).

Traffic accidents are a major public safety issue in Hong Kong, and predicting their severity is important for improving road safety, infrastructure design, and driver education. Statistical models such as logistic regression can help by analyzing factors such as traffic volume, road design, and environmental conditions to understand what contributes to crashes.

This study aims to build reliable models to predict road accident severity in Hong Kong, identify key risk factors, and provide practical recommendations. It will also compare traditional statistical methods, which are easier to interpret, with modern machine learning approaches that may offer higher predictive accuracy, and use these insights to support better transportation planning. At the end of the report, some practical measures are provided for the government's consideration.

2. Dataset

2.1 Data Collection

The data utilized in this study were obtained from the Hong Kong Transport Department. Publicly available datasets (e.g. Road Traffic Accident Database) were downloaded directly from the department's official website ([Road Traffic Accident Statistics - Year 2024](#), 2025), while supplementary information was acquired through formal requests submitted under the Code on Access to Information. The scope of analysis is limited to data from the year 2024 (see Appendix 1).

2.2 Data Preparation

After obtaining the raw accident records, the data were examined to ensure they were suitable for regression analysis. The initial dataset comprised 16,485 accident listing records with 23 variables which covered different aspects of reported road traffic accidents (see Appendix 2).

Due to the relatively small number of fatal and serious accidents in the dataset, these two minority classes were combined into a single category for the analysis. This was done to mitigate issues associated with severe class imbalance, which can negatively affect model estimation and predictive performance.

There were missing values for several variables. We handled the missing values first because incomplete records could affect model reliability. For driver age and vehicle

age variables, the median age of the corresponding vehicle type was used since different vehicle classes tend to have distinct age distributions. It preserved within-group variation and produced more realistic values than using a single overall median. Other categorical variables were imputed based on accidents that occurred within similar time slots.

Data that were not relevant to the scope of the analysis were removed from the dataset. For instance, accidents involving only bicycles were excluded, as the study primarily focuses on car crashes. In addition, observations related to tram, light rail, and other miscellaneous vehicle categories were removed due to the very small number of cases, which could lead to unstable model estimates and unreliable statistical inference.

Furthermore, `light_goods_vehicles`, `medium_goods_vehicles`, and `heavy_goods_vehicles` were combined into a single category because of their similar characteristics and operational functions. This grouping also helps reduce data sparsity and improves model stability and interpretability.

We performed multicollinearity diagnostics on the traffic accident dataset as an additional preprocessing step before regression modeling because regression models would be affected by highly correlated predictors. The Variance Inflation Factor (i.e. VIF) was used to measure how strongly each predictor correlates with the others, as high multicollinearity can destabilize model coefficients. Categorical variables were one-hot encoded, and an initial VIF check was run across the full feature set to identify problematic predictors.

According to the preliminary results, some columns with extreme or infinite VIF scores were dropped. `am_pm` and multiple `time_slot` dummies both return a VIF of infinity and indicate multicollinearity. `am_pm` was removed as these time-related features were redundant with one another. In addition, `road_classification` had a VIF of 174.47 and the `hit_and_run` reached 104.75, both far exceeding the common threshold of 10. They were near-perfectly explained by other predictors in the model. `num_vehicles_involved` also showed a high VIF of 86.60 because it overlapped with `vehicle_type` dummy variables such as `motorcycle`, `taxi`, and `goods_vehicles`.

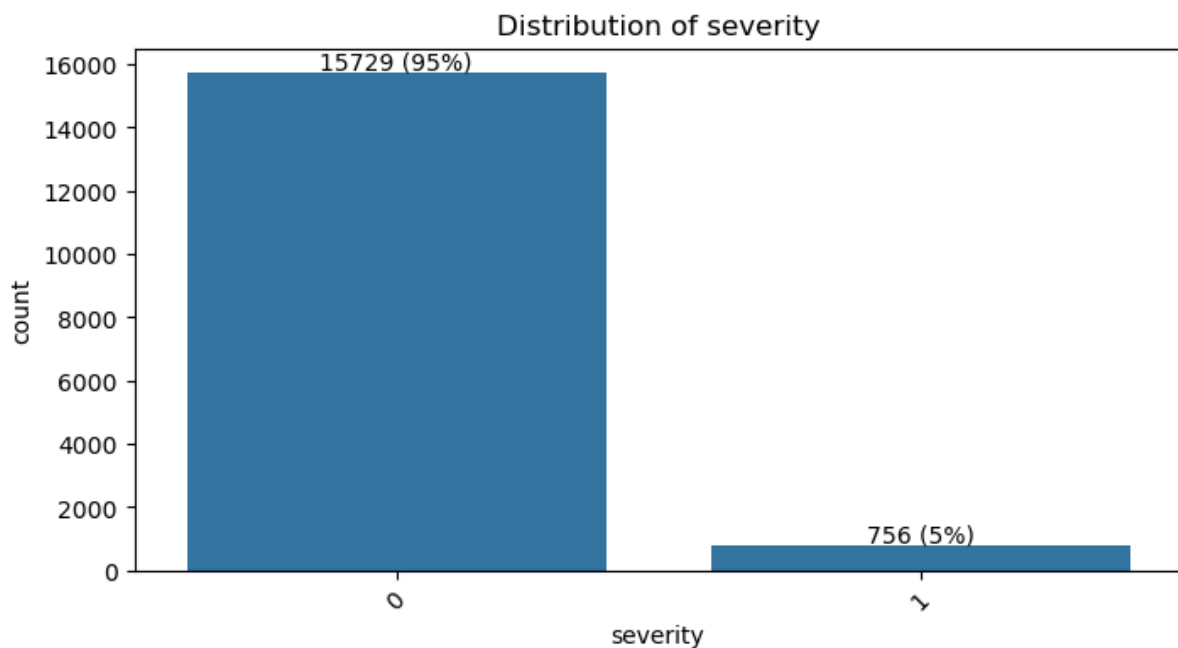
After identifying and removing highly collinear predictors through VIF diagnostics, the remaining variables were retained for modeling because they provide more stable coefficient estimates and clearer interpretation. The table below lists the cleaned 20 variables used in the regression analysis.

Variable Name	Type	Description
severity	Categorical	0 = Slight (One or more persons injured and detention in hospital, if required, will not be more than twelve hours.), 1 = Serious (One or more persons injured and detained in hospital for more than twelve hours, or one or more persons died within 30 days of the accident.)
district	Categorical	WCH, KT, SK, TM, KTS, CW, E, YTM, S, I, KC, WTS, TP, SSPO, ST, N, YL, TW
weather	Categorical	1 = Clear, 2 = Dull, 3 = Fog/mist, 4 = Strong wind
rain	Categorical	1 = Not raining, 2 = Light rain, 3 = Heavy rain
natural_light	Categorical	1 = Daylight, 2 = Dawn/Dusk, 3 = Dark
junction_control	Categorical	1 = No control, 2 = Stop (halt), 3 = Give way (slow), 4 = Traffic signal, 5 = Police, 6 = Not junction
vehicle_movements	Categorical	1 = One moving vehicle, 2 = Two moving vehicles - from same direction, 3 = Two moving vehicles - from opposite direction, 4 = Two moving vehicles - from different roads, 5 = > 2 moving vehicles - from same direction, 6 = > 2 moving vehicles - from opposite direction, 7 = >2 moving vehicles - from different roads
type_of_collision	Categorical	1 = Vehicle collision with Pedestrian, 2 = Vehicle collision with Vehicle, 3 = Vehicle collision with Object (including accidents of unknown accident collision type), 4 = Vehicle without collision
time_slot	Categorical	0 = 0-6, 1 = 6-12, 2 = 12-18, 3 = 18-24
is_holiday	Categorical	0 = Not Holiday, 1 = Holiday
has_young_driver	Categorical	0 = no young driver involved (age < 25), 1 = young driver involved
has_old_driver	Categorical	0 = no old driver involved (age > 65), 1 = old driver involved
has_old_vehicle	Categorical	0 = no old vehicles involved (vehicle age > 10), 1 = old vehicles involved
bicycle	Categorical	0 = accident without bicycle, 1 = accident with bicycle
motorcycle	Categorical	0 = accident without motorcycle, 1 = accident with motorcycle
private_car	Categorical	0 = accident without private car, 1 = accident with private car
public_light_bus	Categorical	0 = accident without public light bus, 1 = accident with public light bus
taxi	Categorical	0 = accident without taxi, 1 = accident with taxi
public_bus	Categorical	0 = accident without public bus, 1 = accident with public bus

goods_vehicle	Categorical	0 = accident without goods_vehicle, 1 = accident with goods_vehicle
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2.3 Univariate Analysis

Most records (95%) fell under severity level 0, while only 5% belonged to severity level 1. A strong class imbalance was identified and affected predictive modeling. This imbalance may pose challenges for predictive modeling as standard algorithms often favor the majority class, potentially leading to poor sensitivity in detecting severity level 1.

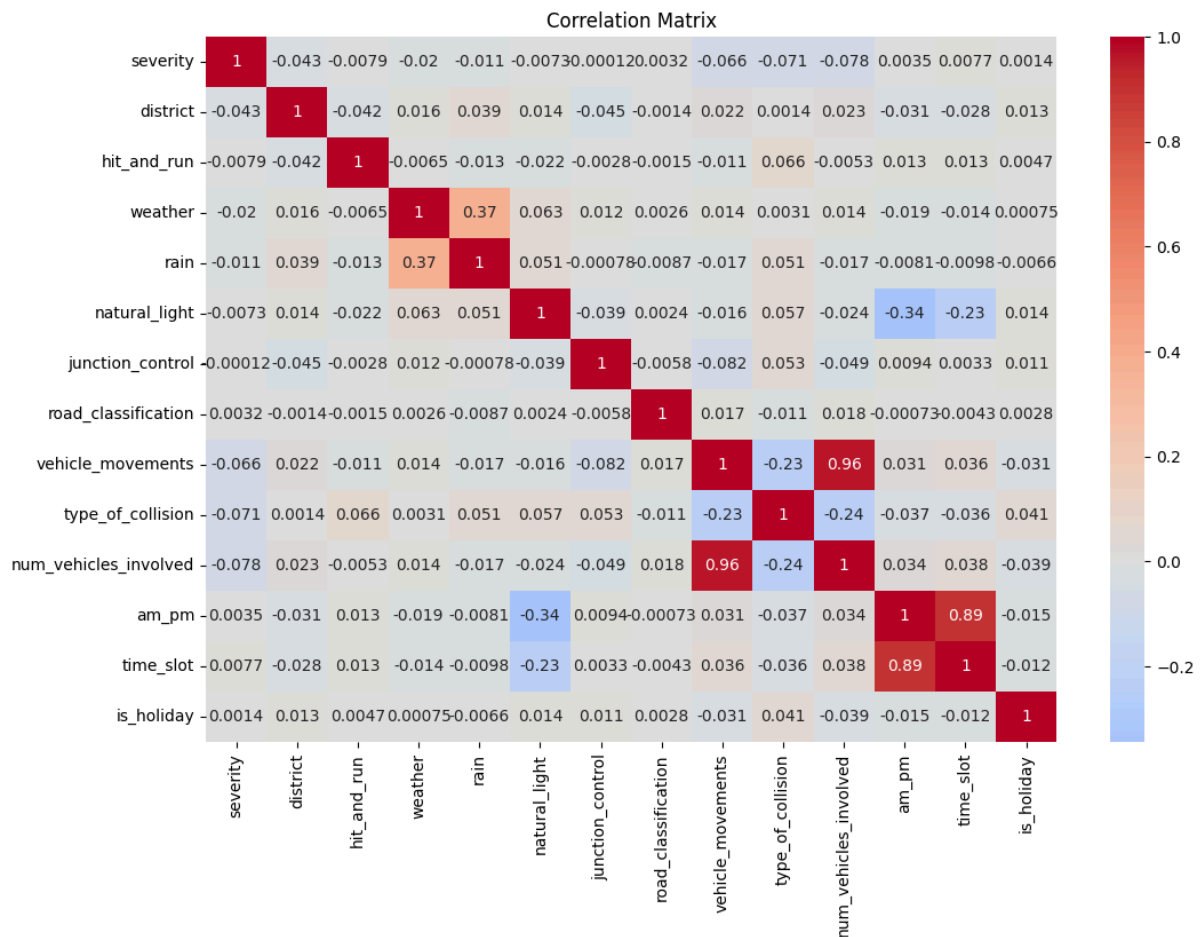


2.4 Bivariate Analysis

The correlation matrix highlighted some important relationships among the variables in the dataset. The strongest correlation observed was between `vehicle_movements` and `num_vehicles_involved` (0.96), indicating that these two variables were almost redundant and may introduce multicollinearity if used together in predictive modeling. Similarly, `am_pm` and `time_slot` showed a very high correlation (0.89) and they encoded overlapping information.

Moderate correlations were found between `natural_light` and time-related variables: it was negatively correlated with `am_pm` (-0.34) and `time_slot` (-0.23). This reflected the intuitive relationship that natural light decreases during nighttime hours. Another notable negative correlation existed between `type_of_collision` and `num_vehicles_involved` (-0.24), which implied that certain collision types tend to involve fewer vehicles.

Limited linear or monotonic relationships when encoded numerically. Most other categorical variables such as `district`, `weather`, `rain`, and `is_holiday` showed very weak correlations with severity. The variable most strongly associated with severity was `num_vehicles_involved`, and it aligned with expectations that accidents involving more vehicles were often more severe.



3. Methodology

Through the above observations, it was evident that certain variables exhibited strong correlations or associations with accident severity, while others showed redundancy or weak influence. To address these issues and build a robust model, we proposed the following methodology.

3.1 Building Logistic Regression Model

Logistic regression would be employed as the primary modelling technique, given that the target variable, severity, was categorical. This method allowed us to estimate the probability of different severity levels based on predictor variables.

3.2 Handling Class Imbalance

The initial dataset revealed an imbalance between severity classes, with the majority class (severity = 0) substantially outnumbering the minority class (severity = 1). To address this issue, we applied the Synthetic Minority Over-sampling Technique (SMOTE) proposed by Nitesh V. Chawla et al. (2002). SMOTE generates synthetic minority class observations by interpolating between existing minority samples and their nearest neighbours, thereby improving minority class representation without simply duplicating observations. Specifically, the minority class was resampled with replacement until it matched the size of the majority class. After oversampling, both classes contained 11,010 records each, resulting in a perfectly balanced dataset.



3.3 Model Performance - Full Model

The logistic regression model was trained and evaluated on the balanced dataset. The performance metrics indicated that the model achieved a reasonable level of predictive accuracy and discrimination between severity classes. The test ROC-AUC score of 0.66 indicated that the model only had modest discriminatory power, and the model's high accuracy was driven by predicting the majority class.


```
Full Model - Confusion Matrix:  
[[4042  677]  
 [ 155   72]]
```

```
Classification Report:
```

	precision	recall	f1-score	support
Slight (0)	0.96	0.86	0.91	4719
Fatal/Serious (1)	0.10	0.32	0.15	227
accuracy			0.83	4946
macro avg	0.53	0.59	0.53	4946
weighted avg	0.92	0.83	0.87	4946

```
Test ROC-AUC : 0.6605
```

3.4 Model Performance - Reduce Model

To improve the model, we applied backward elimination (BE), a step-by-step process that removes variables not meaningfully contributing to predicting accident severity ($p > 0.05$). The selection was performed on training data only to prevent data leakage. Before this process, five low-signal variables including `road_classification`, `hit_and_run`, `weather`, `rain`, and `is_holiday` were removed upfront. After elimination, the final model retained variables related to district, type of collision, vehicles involved, and driver and vehicle characteristics.

Since severe accidents were far less common in the dataset than slight ones, we used a technique called SMOTE to artificially generate more examples of the minority class during training, helping the model learn to detect them better. Features were also standardized before training to ensure fair comparison across variables. The model's performance was tested using 5-fold cross-validation, where the data was split into five parts and the model was trained and evaluated five times to get a reliable performance estimate.

The final model performed well overall. It could identify 3,689 slight accident cases and manage to catch 105 out of the severe cases. 46% of all fatal or serious accidents in the test set were detected. Catching severe accidents was especially important in road safety contexts, even if it came at the cost of some false alarms. The overall accuracy was 77%, and the ROC-AUC score of 0.71 showed the model had a reasonable ability to distinguish between slight and severe accidents. This made it a solid starting point for building more advanced accident severity prediction tools.

```

Reduced Model (Backward Elim) - Confusion Matrix:
[[3689 1030]
 [ 122  105]]

```

Classification Report:

	precision	recall	f1-score	support
Slight (0)	0.97	0.78	0.86	4719
Fatal/Serious (1)	0.09	0.46	0.15	227
accuracy			0.77	4946
macro avg	0.53	0.62	0.51	4946
weighted avg	0.93	0.77	0.83	4946

Test ROC-AUC: 0.7070

After backward elimination, random oversampling (ROS) was used to duplicate existing severe accident cases to increase the representation of the minority class during training. The ensemble method further combined multiple models trained on different resampled subsets of the data to improve robustness and reduce instability associated with individual resampling methods.

3.5 Model Comparison

The full model achieved the highest accuracy (0.8318) and F1 score (0.5271). However, it had the lowest ROC-AUC among the four models. This suggested a trade-off between overall accuracy and the model's ability to discriminate and identify severe cases. In particular, the relatively low recall indicated that many severe accidents were not being detected.

Although random oversampling (ROS) and ensemble methods achieved slightly lower overall accuracy, they yielded higher recall for severe accidents, indicating a stronger ability to correctly identify high-risk cases.

A single model could be sensitive to sampling noise, which would make unstable predictions. Resampling techniques aimed to address class imbalance but they had their own limitations. For instance, SMOTE may introduce noise since new synthetic instances were created by interpolating between nearest neighbours, which might not be appropriate for a dataset with many categorical variables. Similarly, random oversampling duplicated existing severe outcomes which may lead to overfitting of the model.

To address these limitations, the ensemble approach combined multiple models trained on different resampled datasets. The ensemble model enhanced performance by reducing the instability associated with resampling and capturing a broader range of patterns in the dataset. As a result, it provided more robust and generalizable predictions, particularly in identifying severe accidents. This finding

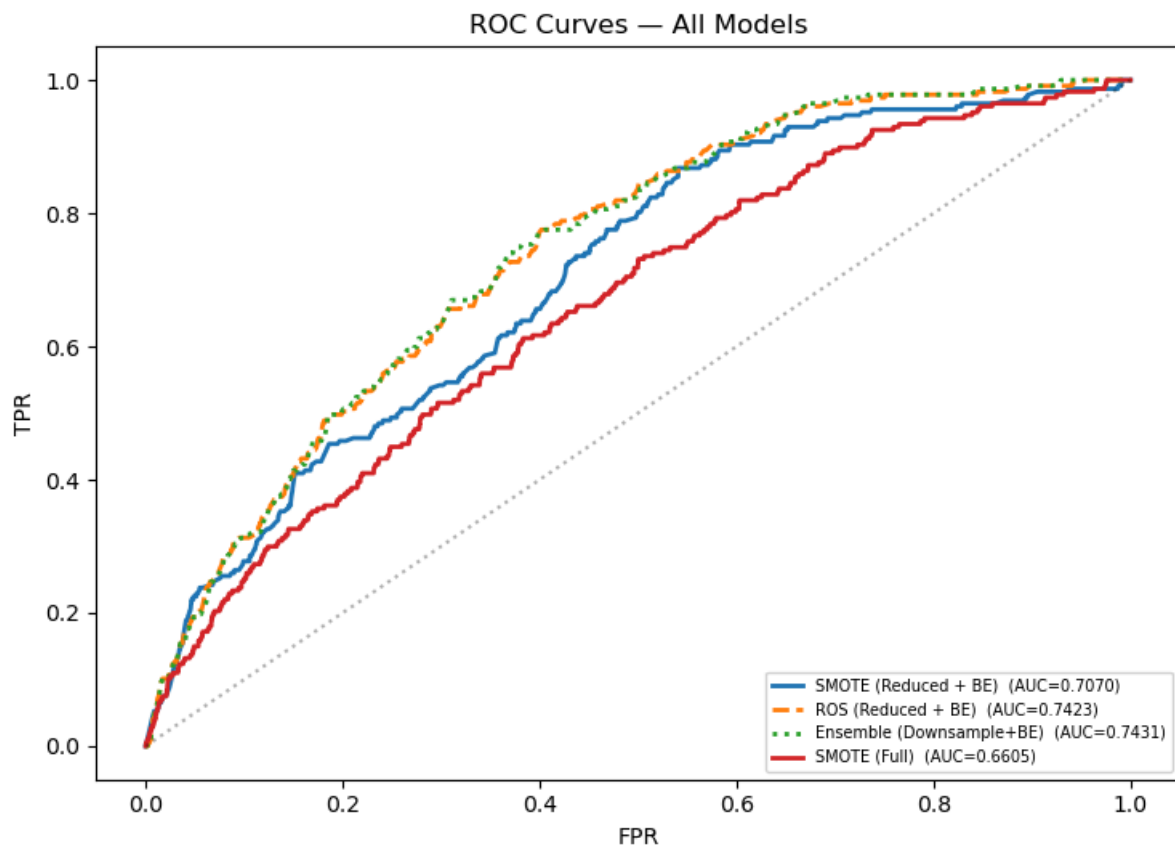
was further supported by the comparison of ROC-AUC curves, as shown below, where the ensemble model achieved the highest ROC-AUC (0.7431) among the four models.

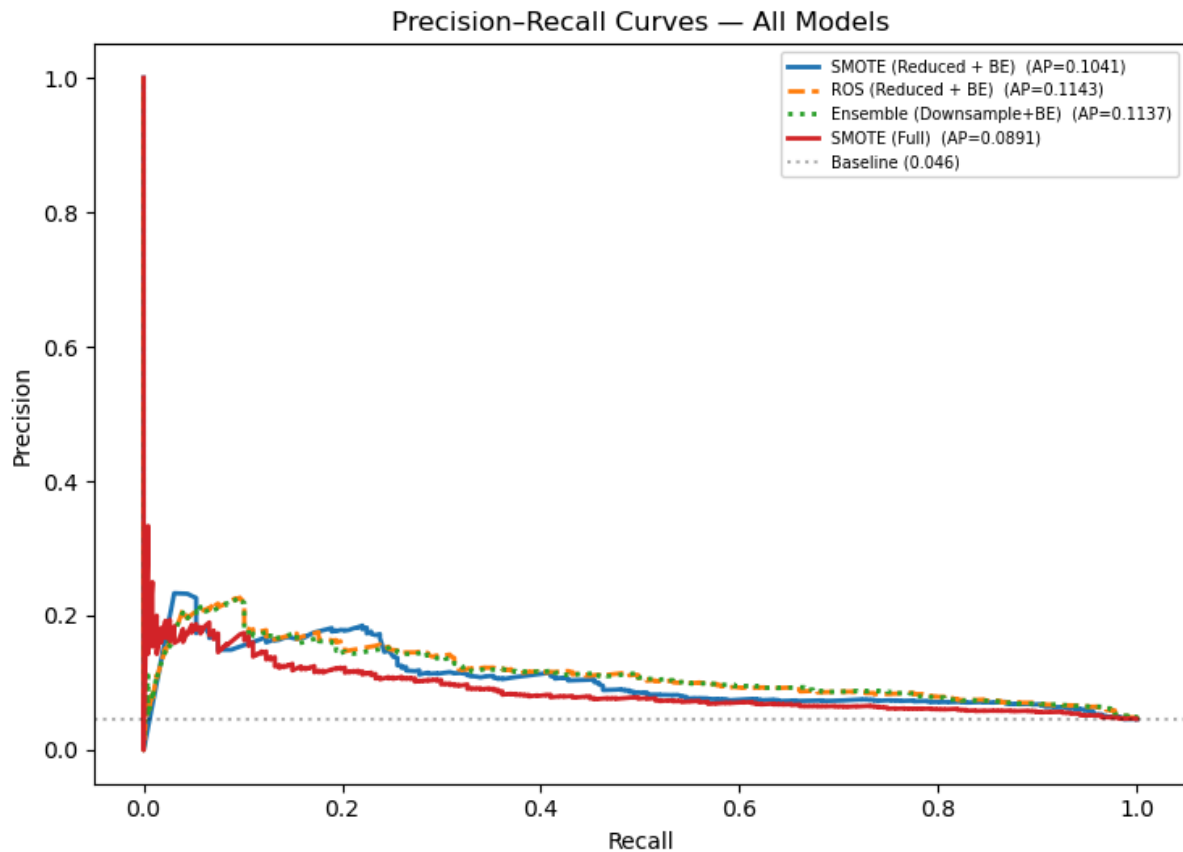
Overall, the ensemble model was preferred for this study, as it offered the best balance between discrimination ability and sensitivity, making it more suitable for identifying severe accidents where minimizing false negatives was of primary importance.

4. Results

After the model comparison, the results are as follows.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Full Model (SMOTE)	0.8318	0.0961	0.3172	0.5271	0.6605
Reduced Model (SMOTE + BE)	0.7671	0.0925	0.4626	0.5096	0.707
Reduced Model (ROS + BE)	0.7056	0.0922	0.6123	0.4909	0.7423
Ensemble (100 LR, downsample + BE)	0.7	0.0937	0.6388	0.4903	0.7431





We based the reduced model on our final model and converted the coefficients (β) into Odds Ratios. The Odds Ratio let us understand how many times the odds increase or decrease for a one-unit change in the predictor. The conversion formula was $OR = e^{\beta}$.

Category	Variable	Odds Ratio	Interpretation
Vehicle Factors (Accident with Vehicle type)	Motorcycle	1.42	Higher severity
	Bicycle	1.11	Slightly higher severity
	Public bus	0.92	Slightly lower severity
	Goods vehicle	0.92	Slightly lower severity
	Old vehicle	0.96	Minimal effect
Weather (compare with the weather is "Clear")	Fog/mist	0.96	Minimal effect
	Strong wind	0.90	Lower severity odds
	Dull weather	0.78	Lower severity odds
Rain (compare with "Not raining")	Light rain	0.84	Lower severity odds
	Heavy rain	0.75	Substantially lower odds of severity

Vehicle Movements (compare with "One moving vehicle")	Two moving vehicles - from opposite directions	0.77	Lower severity odds
	Two moving vehicles - from the same direction	0.52	Significantly lower severity odds
Collision types (compare with "Vehicle collision with pedestrian")	Vehicle collision with vehicle	0.58	Less severe than pedestrian collisions
	Vehicle collision with object	0.60	Lower severity
	Vehicle without collision	0.45	Much lower severity

Motorcycle involvement was associated with higher crash severity, whereas several adverse weather and multi-vehicle movement conditions were associated with lower observed severity odds, possibly reflecting behavioral adaptation and reduced exposure rather than inherently safer conditions.

It is worth noting that heavy rain had a lower odds ratio (0.75) than light rain (0.84). It should not be over-interpreted that driving in heavy rain is safer. A developed risk compensation theory argues that drivers adjust their behavior including alertness, caution, and speed to maintain a target level of risk. ("Does Risk Homeostasis Theory Have Implications for Road Safety," 2002). During a downpour, most drivers naturally slow down to compensate for poor visibility and more challenging road conditions, engaging in a form of self-correction that can reduce crash risk.

Drivers are often more focused and alert because the danger is visible and immediate, which heightens their vigilance and encourages more cautious behavior. A study demonstrates that driver alertness is maintained during simulated rain, supported by high beta wave activity, while fast and hasty driving behaviours decrease (Susanto et al., 2025) .

Apart from the driver behavior, vulnerability and exposure of road users may interpret the odds ratio. Good weather often increases the sheer volume of vulnerable users interacting with fast-moving traffic (Sudhakaran et al., 2025). More cyclists and motorcyclists tend to be on the road, which increases the exposure of these vulnerable road users to potential traffic conflicts. During bad weather, many of these individuals choose to stay indoors or avoid travel and reduce their presence on the road, potentially causing fewer total casualties among these specific groups (VE-I-4: Road Accidents Due to Weather Conditions and Weather Patterns, 2023).

5. Conclusion

The study confirms that logistic regression, when applied to a carefully selected set of variables, provides a robust and interpretable framework for predicting accident severity. By simplifying the dataset through regrouping and exclusion of imbalanced predictors, the reduced model maintained accuracy while achieving a higher ROC-AUC score, demonstrating improved stability and discriminatory power. Key drivers such as vehicle involvement, collision type, and district characteristics highlight meaningful risk factors, while the removal of low-value variables enhances clarity. Overall, the reduced model balances statistical rigor with practical interpretability, offering actionable insights for traffic safety policy, resource allocation, and public communication. It establishes a scalable foundation for ongoing monitoring and refinement of accident severity prediction. The future research will focus on developing hybrid approaches, machine learning, and neural networks to enhance predictive performance.

6. Way forward

The analysis highlights several key factors that significantly impact accident severity, namely the involvement of vulnerable road users, weather and road surface conditions, specific district, and single-vehicle collisions. Based on these findings, we propose the following measures to reduce the risk of severe outcomes for Hong Kong government's consideration:

1. Prioritize Protection for Vulnerable Road Users

It should expand pedestrian priority zones, install traffic calming measures like speed bumps at crossings, and plan more dedicated lanes for motorcycles and bicycles. Strictly enforce regulations requiring drivers to yield at pedestrian crossings and ensure that two-wheeler riders wear approved safety equipment

2. Promote Attentive Driving and Post-Weather Vigilance

Educating drivers that emphasizing good visibility often leads to fatal speeding and urging drivers to maintain focus at all times. The government should remind the public to be aware about road conditions following severe weather like heavy rain or typhoons, particularly slippery surfaces to prevent accidents caused by skidding or loss of vehicle control.

3. Target High-Risk Districts

It should optimize the deployment locations of ambulances within high-risk zones to shorten the rescue time and ensure rapid emergency medical response when severe accidents occur. Investigative and re-design the roading plan within the high-risk districts (e.g. Central Western district).

4. Enhance Infrastructure for Single-Vehicle Accidents

It should review and upgrade road safety infrastructure like installing advanced crash barriers and energy-absorbing crash cushions along high-risk curves to reduce impact to avoid severe injuries in single-vehicle accidents such as self-collision.

7. References

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<https://doi.org/10.18280/ijssse.150816>

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<https://www.umweltbundesamt.de/en/themen/klima-energie/klimafolgen-anpassung/folgen-des-klimawandels/monitoring-zur-das/das-handlungsfelder-indikatoren/verkehr-verkehrsinfrastruktur/ve-i-4-road-accidents-due-to-weather-conditions>

Appendix

Appendix 1. Email Request of Transport Department under the Code on Access to Information

2026/4/11 上午11:32

Mail - KWOK, Kai Yin - Outlook

 Outlook

Fw: Request for Access to Information (No. 108/26)

From YAU, Chun Wing <1155251820@link.cuhk.edu.hk>
Date Thu 2026-03-19 14:00
To KWOK, Kai Yin <1155251560@link.cuhk.edu.hk>; NG, Ching Wai <1155253020@link.cuhk.edu.hk>; HO, Ka Ho <1155246984@link.cuhk.edu.hk>; CHAN, Chi Chiu <1155253025@link.cuhk.edu.hk>; TANG, Siu Wan <cuhkewantang@link.cuhk.edu.hk>

 2 attachments (2 MB)
code table.xls; Attachment2024.xls;

From: Natalie Ho Yau FUNG <nataliehoyaufung@td.gov.hk>
Sent: Thursday, March 19, 2026 12:59:58 PM
To: YAU, Chun Wing <1155251820@link.cuhk.edu.hk>
Subject: Request for Access to Information (No. 108/26)

Dear Mr YAU,

I refer to your request for access to information no. 108/26 about information on Road Traffic Accident Statistics for the year 2024.

Your requested data of traffic accidents happened in the year of 2024 (from 1 January 2024 to 31 December 2024) are attached.

Note:
Number "8" shall be added as prefix for the number sets under the columns "Coordinates of accident (easting grid)" and "Coordinates of accident (northing grid)" on the worksheet of the attached excel file to arrive at the "Easting" and "Northing" coordinate of the Hong Kong Metric Grid 1980.

Yours sincerely,
Miss Natalie FUNG
for Commissioner for Transport

From: "YAU, Chun Wing" <1155251820@link.cuhk.edu.hk>
To: "tdenq@td.gov.hk" <tdenq@td.gov.hk>
Cc: "KWOK, Kai Yin" <1155251560@link.cuhk.edu.hk>; "HO, Ka Ho" <1155246984@link.cuhk.edu.hk>; "CHAN, Chi Chiu" <1155253025@link.cuhk.edu.hk>; "NG, Ching Wai" <1155253020@link.cuhk.edu.hk>; "TANG, Siu Wan" <cuhkewantang@link.cuhk.edu.hk>
Date: 11/03/2026 11:34 PM

<https://outlook.office.com/mail/inbox/id/AAQkADgwNmVINmRjLWJkYzktNDZKYi11YT13LTk3OTFkOTcwNjhjMwAQAAE%2FCiI2weFJi1GbX7Y%2B...> 1/2

2026/4/11 上午11:32

Mail - KWOK, Kai Yin - Outlook

Subject: Request under Code on Access to Information - Traffic Accident Data of Hong Kong by Incident Level

Dear Access to Information Officer,

I am writing to make a formal request for information under the Code on Access to Information.

We are a group of university students conducting a data science project analysing patterns and contributing factors in road traffic accidents in Hong Kong for academic purposes. Although there's a Road Traffic Accident Database 2024 on the website of Transport Department, some details such as date and time are not available for download. As part of this project, we would like to request detailed but anonymized traffic accident records for the calendar year 2024 at the incident level.

Specifically, we kindly request the following traffic accident data for the period from 1 January 2024 to 31 December 2024:

- Serial number of the accident
- Date of accident
- Time of accident (or time range/hour if exact time is not releasable)
- Location including district, road name, latitude, and longitude

Please provide the information in electronic format (Excel, CSV, or similar) if possible.

Thank you very much for your assistance.

Yours faithfully,
Ryan Yau

Appendix 2. Initial Dataset from Transport Department

Variable Name	Type	Description
severity	Categorical	0 = Slight (One or more persons injured and detention in hospital, if required, will not be more than twelve hours.), 1 = Serious (One or more persons injured and detained in hospital for more than twelve hours, or one or more persons died within 30 days of the accident.)
district	Categorical	WCH, KT, SK, TM, KTS, CW, E, YTM, S, I, KC, WTS, TP, SSPO, ST, N, YL, TW
hit_and_run	Categorical	1 = Yes, 2 = No
weather	Categorical	1 = Clear, 2 = Dull, 3 = Fog/mist, 4 = Strong wind
rain	Categorical	1 = Not raining, 2 = Light rain, 3 = Heavy rain
natural_light	Categorical	1 = Daylight, 2 = Dawn/Dusk, 3 = Dark
junction_control	Categorical	1 = No control, 2 = Stop (halt), 3 = Give way (slow), 4 = Traffic signal, 5 = Police, 6 = Not junction
vehicle_movements	Categorical	1 = One moving vehicle, 2 = Two moving vehicles - from same direction, 3 = Two moving vehicles - from opposite direction, 4 = Two moving vehicles - from different roads, 5 = > 2 moving vehicles - from same direction, 6 = > 2 moving vehicles - from opposite direction, 7 = >2 moving vehicles - from different roads
type_of_collision	Categorical	1 = Vehicle collision with Pedestrian, 2 = Vehicle collision with Vehicle, 3 = Vehicle collision with Object (including accidents of unknown accident collision type), 4 = Vehicle without collision
num_vehicles_involved	Numeric	No. of Vehicles Involved
am_pm	Categorical	0 = AM, 1 = PM
time_slot	Categorical	0 = 0-6, 1 = 6-12, 2 = 12-18, 3 = 18-24
is_holiday	Categorical	0 = Not Holiday, 1 = Holiday
has_young_driver	Categorical	0 = no young driver involved (age < 25), 1 = young driver involved
has_old_driver	Categorical	0 = no old driver involved (age > 65), 1 = old driver involved
has_old_vehicle	Categorical	0 = no old vehicles involved (vehicle age > 10), 1 = old vehicles involved
bicycle	Categorical	0 = accident without bicycle, 1 = accident with bicycle
motorcycle	Categorical	0 = accident without motorcycle, 1 = accident with motorcycle
private_car	Categorical	0 = accident without private car, 1 = accident with private car

public_light_bus	Categorical	0 = accident without public light bus, 1 = accident with public light bus
taxi	Categorical	0 = accident without taxi, 1 = accident with taxi
public_bus	Categorical	0 = accident without public bus, 1 = accident with public bus
goods_vehicle	Categorical	0 = accident without goods_vehicle, 1 = accident with goods_vehicle